PHYSICAL CHEMISTRY

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Topic : Chemical Equilibrium & Ionic Equilibrium (Question Bank # 02)

MM : 64 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. A buffer solution was prepared by dissolving 0.02 mole acetic acid & 0.01 mole sodium acetate in enough water to make 1.0 L of solution $\left(K_{a(CH_3COOH)} = 2\times10^{-5}\right)$. If 5×10^{-4} mole NaOH were added to 100 ml of the buffer. The resultant pH of the solution would be [given log 5 = 0.70]

(A) 5 (B) 4.7 (C) 5.3 (D) None

2. PCl_5 (molecular mass = M) dissociates into PCl_3 and Cl_2 as : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. If the total pressure of the system at equilibrium is P and the density is 'd' at temeprature , T K. The degree of dissociation of PCl_5 may be represented as

(A) $\frac{PM}{dRT}$ (B) $\frac{PM}{dRT}-1$ (C) $\frac{dRT}{PM}-1$ (D) $\frac{dRT}{PM}$

3. Solid Ammonium carbamate dissociates as: NH_2COONH_4 (s) $\rightleftharpoons$ $2NH_3(g) + CO_2(g)$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.

(A) $\frac{31}{9}$ (B) $\frac{31}{27}$ (C) $\frac{27}{31}$ (D) $\frac{9}{31}$

4. Which of the following can not act as buffer solution -

(A) 30 ml 0.1M CH_3COOH + 10 ml 0.3M NH_4OH

(B) 10 ml 0.2M Na_2CO_3 + 20 ml 0.2M H_2CO_3

(C) 30 ml 0.1M HCl + 10 ml 0.1M Na_3PO_4

(D) 10 ml 0.1M NH_4Cl + 5 ml 0.1M NaOH

5. A certain gas 'A' polymerizes to a very small extent at a given temperature as : $nA(g) \rightleftharpoons A_n(g)$. The reaction is started with one mole of 'A' in a container of capacity V. Which of the following is correct value of $\frac{PV}{RT}$, at equilibrium?

(A) $1-\frac{(n-1).K_C}{V^{n-1}}$ (B) $\frac{(n-1).K_C}{V^{n-1}}$ (C) $1-\frac{n.K_C}{V^{n-1}}$ (D) $1-\frac{n.K_C}{V^{n-1}}$

6. The increasing order of Ag^+ ion concentration in :

(I) Saturated solution of AgCl (II) Saturated solution of AgI

(III) 1M $Ag(NH_3)_2^+$ in 0.1 M NH_3 (IV) 1M $Ag(CN)_2^-$ in 0.1 M KCN

Given : $K_{sp}(AgCl) = 10^{-10}$; $K_{sp}(AgI) = 10^{-16}$, $K_f(Ag(NH_3)_2^+) = 10^8$; $K_f(Ag(CN)_2^-) = 10^{11}$

(A) II < I < III < IV (B) IV < III < II < I

(C) IV < II < III < I (D) IV < II < I < III

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7. pH of a solution by mixing 50 mL of 0.1 M Na_3PO_4 and 50 mL of 0.05 M NaH_2PO_4 is :
[For H_3PO_4 : $pK_1 = 2.1$, $pK_2 = 7.2$, $pK_3 = 12$ and log 3 = 0.3]

(A) 9.6 (B) 4.65 (C) 7.2 (D) 11.7

8. For the reaction

$[Ag(CN)_2]^- \rightleftharpoons Ag^+ + 2CN^-$; K_c at 25°C is 4×10^{-19}

The equilibrium concentration of Ag^+ is, in a solution which is 0.1 M in KCN & 0.03 M in $AgNO_3$ originally.

(A) 7.5×10^{-16}M (B) 7.5×10^{-18} M (C) 1.25×10^{19}M (D) 1.25×10^{-17}M

9. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{th}$ of that of a strong acid (HX, 1M), at 25°C. The K_a of HA is

(A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

(ii) One or more options correct Type

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

10. A reaction at 300K with $\Delta G^\circ = -1743$ J consists of 3 moles of A(g), 6 moles of B(g) and 3 moles of C(g). If A, B and C are in equilibrium in 1 L container, then the reaction may be.
(ln 2 = 0.7, R = 8.3 J/K-mol).

(A) $A + B \rightleftharpoons C$ (B) $A \rightleftharpoons B + 2C$ (C) $2A \rightleftharpoons B + C$ (D) $A + B \rightleftharpoons 2C$

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11. Which of the following combination of solutes would result in the formation of a buffer solution ?

(A) $NH_3 + NH_4Cl$ (B) $CH_3COOH + NaOH$ in 2 : 1 molar ratio

(C) $NH_3 + HCl$ in 2 : 1 molar ration (D) $CH_3COOH + NaOH$ in 1 : 2 molar ratio

12. Choose the correct alternatives

(A) K_{sp} of $Fe(OH)_3$ in aqueous solution is 3.8×10^{-38} at 298 K. The concentration of Fe^{3+} will increases with decrease in H^+ ion concentration.

(B) In a mixture of NH_4Cl and NH_4OH in water, a further amount of NH_4Cl is added. The pH of mixture will decrease.

(C) An aqueous solution of each of the following salts NH_4I, $Cr(NO_3)_3$, KCN will be basic, acidic and neutral respectively.

(D) Degree of hydrolysis of 0.1 M CH_3COONH_4, 0.2 M CH_3COONH_4, 0.5 M CH_3COONH_4 is same.

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(iv) Table Type

- This section contains **3** questions of matching type.
- This section contains **ONE** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

Column-I Compound	Column-II Relationship	Column-III Statement
(A) CH_3COOH ($K_a = 2 \times 10^{-5}$)	(P) $pH = \frac{1}{2}(P^{Ka_1} + P^{Ka_2})$	(I) Addition of HCl(aq) can make buffer solution
(B) 0.1 M $NaHCO_3$	(Q) $pH = \frac{1}{2}(pK_a - \log C)$	(II) Increases solubility of AgCl(s)
(C) NH_3(aq.) ($K_b = 2 \times 10^{-5}$)	(R) $pH = 7 + \frac{1}{2}(pK_b + \log C)$	(III) pH is independent of concentration
(D) CH_3COONa(aq.)	(S) $pOH = \frac{1}{2}(pK_b - \log C)$	(IV) Addition of NaOH (aq.) can make buffer solution

13. Which of the following option is incorrectly matched -
(A) A → Q → IV (B) B → S → IV (C) C → S → II (D) C → S → I

14. 30 ml of 0.1 M CH_3COOH (aq.) is mixed with 10 ml of 0.1 M NaOH(aq.) pH of resulting solution
(A) 4.7 (B) 5.3 (C) 4.4 (D) 5.6

15. For H_2CO_3 ($K_{a_1} = 10^{-7}$; $K_{a_1} = 10^{-11}$)
Number of OH^- in 1 litre of 0.1 M $NaHCO_3$(aq.) is -
(A) $10^{-5} \times N_A$ (B) $10^{-9} \times N_A$ (C) $10^{-7} \times N_A$ (D) $10^{-11} \times N_A$

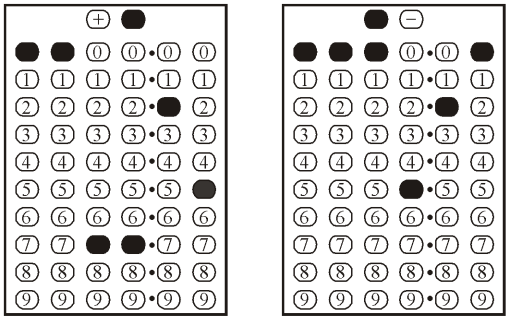
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SECTION-II: (Integer Second Decimal)

- This section contains **4** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

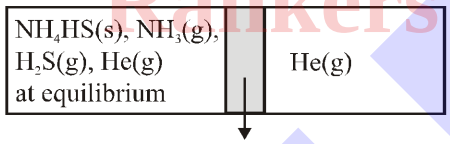
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1. 10 ml of 0.1M weak acid HA($k_a = 10^{-5}$) is mixed with 10 ml 0.2M HCl and 10 ml 0.1M NaOH. If the value of $[A^-]$ in the resulting solution is $x \times 10^{-5}$ M the value of 'x' is.

2 The solubility of $Pb(OH)_2$ in water is 2.0×10^{-6}M. If the solubility of $Pb(OH)_2$ in a buffer solution of pH = 8 is $x \times 10^{-5}$ M the value of 'x' is.

If your answer is $x \times 10^{-y}$ then fill 'y' in OMR.

3.



Fixed SPM, which allows only He(g) to cross it

The entire system is at equilibrium at 300 K. The volume of each chamber is 82.1 L. The total pressure in left chamber is 4 atm and in right chamber is 2 atm. $NH_3(g)$ and $H_2S(g)$ are obtained only from the dissociation of $NH_4HS(s)$. The value of K_P (in atm^2) for the reaction: $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, is

4. $COF_2(g)$ passed over catalyst at 1000°C comes to equilibrium: $2COF_2(g) \rightleftharpoons CO_2(g) + CF_4(g)$. Analysis of the equilibrium mixture (after quick cooling to freeze the equilibrium) shows that 500 ml of the equilibrium mixture (STP) contains 300 ml (STP) of (COF_2 + CO_2) taking the total pressure to be 10 atm. The value of K_P for the reaction is

ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1	**Ans.(B)**	**2. Ans.(B)**	**3. Ans.(B)**	**4. Ans.(C)**	**5. Ans.(A)**
6.	**Ans.(C)**	**7. Ans.(D**	**8. Ans.**(B)	**9. Ans.**(A)	

(ii) One or more options correct Type : 4(–1)

10.	**Ans.(C)**	**11. Ans.(A,B,C)**	**12. Ans.(B,D)**

(iv) Table Type : 3(0)

13.	**Ans.(B)**	**14. Ans.(C)**	**15. Ans.(A)**

SECTION-II: (Integer Second Decimal) : 4(0

1.	**Ans. (1.00)**	**2. Ans.(3.20)**	**3. Ans. (1.00)**	**4. Ans.(4.00)**

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